

MotionSense Gait Dynamics: Signal-Based Analysis using RNN and Dimensionality Reduction

Abstract

This project analyzes human gait from smartphone IMU downstairs-walking data as a continuous time series rather than a classification task. Triaxial user acceleration from subjects of different ages and genders is modeled using two methods taught in the course: a SimpleRNN for next-step prediction and PCA for low-dimensional representation. The RNN, trained on youth females and evaluated on youth females and early-adult males, uses per-subject prediction error as an unsupervised similarity measure and shows lower error for the training group, indicating group-specific dynamics. PCA, applied separately to each age-gender group, finds a common rhythmic structure but different x/y/z contributions, demonstrating that time-series techniques can reveal demographic structure in IMU gait data.

Background and Problem Statement

Smartphone IMUs yield a rich time-series measurement of gait. Instead of classifying activities, this work models gait as a dynamical process and explores how well a sequence predictor, trained on one demographic group, generalizes to another. A SimpleRNN that is best at predicting youth-female gait does indicate a higher prediction error on the data from the early-adult male group, suggesting demographic differences in gait dynamics. PCA sheds insight into such differences: in comparison with the signals of youth and early-adult males, the acceleration signals of youth females have sharper, higher-frequency vertical motion. Such low-dimensional PCA patterns provide an interpretation for why a female-trained predictor does not generalize perfectly and link prediction error directly to the biomechanical differences between populations.

Dataset Description

The notebook uses a MotionSense dataset with the following properties:

- **Subjects:** 24 smartphone users, each with metadata: subject, code, age, gender, weight, and height.
- **Activities:** Several labeled activities; this project uses only **downstairs walking** (dws).
- **Signals:** DeviceMotion userAcceleration.x, userAcceleration.y, and userAcceleration.z as a 3D time series sampled at a fixed rate.

Subjects are grouped by:

- **Age:** youth (≤ 25), early-adult, and late-adult.
- **Gender:** female (0) and male (1).

For the **RNN experiment**:

- **Training group:** youth females (four subjects). Pooled downstairs-walking data have shape approximately $20,600 \times 3$.
- **Test group:** early-adult males (six subjects), with approximately $32,800 \times 3$ total samples.

For the **PCA experiment**, separate datasets are constructed for youth male, youth female, early-adult male, early-adult female, and late-adult male groups, each consisting of all subjects' 3D downstairs-walking acceleration in that group.

Methods

1. RNN Prediction Error

Preprocessing: For each group, all downstairs-walking segments are concatenated into a single 3D sequence, with subject index ranges stored; the pooled youth-female signal is then standardized per axis. A Keras SimpleRNN is trained as a one-step-ahead predictor using 64-sample sliding windows (SimpleRNN(16, tanh) followed by a 3-unit dense layer, MSE loss, 10 epochs with TimeseriesGenerator). This model learns short-term gait dynamics for the youth female training group. For evaluation, each subject's 3D signal is normalized with the training-group statistics, segmented into 64-step windows, and used to predict the 65th sample; per-time-step MSE is computed and summarized by mean and standard deviation per subject. These mean MSE values are then compared between youth females and early-adult males and displayed in a per-subject bar chart.

What one training example looks like

You pick a **window length** $L = 64$.

For a given time index t , one training example is:

Input window (past 64 samples):

$$X(t), X(t+1), \dots, X(t+63)$$

This is a little time-chunk:

$$\begin{bmatrix} x(t) & y(t) & z(t) \\ x(t+1) & y(t+1) & z(t+1) \\ \vdots & \vdots & \vdots \\ x(t+63) & y(t+63) & z(t+63) \end{bmatrix}$$

where:

- $x(t)$ = userAcceleration.x at time t
- $y(t)$ = userAcceleration.y at time t
- $z(t)$ = userAcceleration.z at time t

Target (what we want the network to predict):

$$X(t + 64) = \begin{bmatrix} x(t + 64) \\ y(t + 64) \\ z(t + 64) \end{bmatrix} \text{shape } (3,)$$

So intuitively:

“Look at how x, y, z evolved over the last 64 timesteps, and guess what the next x, y, z will be.”

What the SimpleRNN does with that window

The SimpleRNN takes the **sequence over time**:

$$X(t), X(t + 1), \dots, X(t + 63)$$

At each time k in the window, the RNN updates its hidden state:

$$h(k) = \tanh(W_x X(k) + W_h h(k - 1) + b)$$

- $X(k)$ is 3D: $[x(k), y(k), z(k)]^T$
- $h(k)$ is, say, 16D (hidden units)
- W_x maps the 3 input axes into the hidden space
- W_h carries the “memory” from the previous time

After processing the whole window up to $t + 63$, you end up with a **final hidden state**:

$$h_{\text{final}} = h(t + 63) \text{shape } (16,)$$

Intuition:

- The RNN has watched the movie of x, y, z from time t to $t + 63$.
- h_{final} is its compressed memory of the movement pattern in that window (stride pattern, amplitude, direction changes, etc.).

2. PCA by Age-Gender Group

For each age-gender group:

1. All subjects' 3D downstairs-walking signals are concatenated into a matrix $X \in \mathbb{R}^{T \times 3}$.
2. PCA with **one component** is applied:

$$X \approx z w^T, w \in \mathbb{R}^3, \|w\| = 1$$

where z is a 1D time series (PC1) and w is the loading vector indicating how much each axis contributes.

3. The PC1 loadings (w_x, w_y, w_z) are recorded for each group.
4. The absolute values of the loadings are plotted to show which axes dominate the main gait direction in each demographic group.

This yields a compact linear representation of gait for each age-gender group and allows direct comparison of axis contributions.

Results

• RNN Prediction Error

For youth female subjects (training group), most mean MSE values are low (≈ 0.2 – 0.4), with one clear high-error outlier; excluding this outlier, the group is predicted well by the model. Early-adult male subjects (test group) show consistently higher mean MSEs (≈ 0.6 – 0.9), giving a larger group-average error than youth females. The per-subject bar chart therefore indicates that, even without label information, prediction error alone carries demographic information: a model trained on youth female gait yields higher errors for early-adult male gait.

• PCA Loadings

PC1 loadings show that age-gender groups differ in how x, y, and z contribute to the main gait direction. Youth males and late-adult males are dominated by y and z (with little x for late-adult males), youth females emphasize x and z, and early-adult males and females have more balanced contributions from all three axes. The PC1 time series is quasi-periodic for all groups, but the distribution of motion across lateral (x), forward–backward (y), and vertical (z) directions varies systematically with age and gender.

Discussion and Conclusion

This demonstrates that a SimpleRNN, trained on youth female gait, captures short-term patterns well within the same demographic but generalizes poorly to other groups. The higher prediction error on early-adult male subjects suggests that the learned dynamics are demographic-specific rather than universal gait rules. PCA helps explain why: compressing the 3-axis signal into one principal direction reveals clear differences in axis contributions across age-gender groups, implying that each group exhibits distinct gait structure. Thus, PCA serves not as a standalone method but as evidence for why the SimpleRNN fails to predict across demographics; because the underlying gait manifolds differ. While the dataset and models are simple, the approach demonstrates how timeseries prediction and dimensionality analysis can jointly uncover demographic structure in gait and motivates the need for richer, nonlinear models to capture cross-group variability.